Metal Ledger & Valuation Engine

Day 1 Design Note · October 13, 2026

A rate is locked **per line**, the moment that line is added to the bill — not per bill, not at print time.

1. The Day 1 question

Each line keeps the rate that was live when it was added. A bill that spans a rate change can legitimately show two different rates on two of its lines.

- Item 1 (10:02) → keeps Rate A, even after the 10:20 rate change.
- Item 2 (10:25, after the change) → gets Rate B.
- Receipt (10:30) → shows each line's own rate. The customer's total never silently shifts after they agreed to it.
- Old-gold exchange → same rule: locked at the rate live when it was weighed and added.

Enforced with two owner-set limits, checked on the server when the bill is finalized:

- **Quote validity window** — a locked rate only holds for N minutes (default 15). After that, the line must be re-quoted.
- **Rate-change tolerance** — if the live rate has moved more than a set % (default 0.5%) since the line was locked, the cashier is forced to confirm before printing.

2. Data model

- `daily_rates` : add a buy/sell rate type — today there's only one rate, not two.
- Auto-fetch skips a metal+purity already corrected by the Admin that day, instead of overwriting it.
- Gram-per-tola becomes one Settings value, not hardcoded twice in the code.
- `transactions` : add `pure_weight_grams` (stored at write time) and `quoted_at` (the moment the rate was locked).
- Party balance: track **metal owed** and **money owed** as two separate numbers, never combined.
- No new tables — every change extends what already exists.

3. Rounding policy

- Weight → 3 decimal places.
- Money → 2 decimal places.
- Round once, at the moment it's stored — every later calculation reuses that stored number instead of re-rounding a fresh one. This is what guarantees two cashiers reach the same figure.

4. Stretch goals — not built yet, but already fit

- **Karigar reconciliation** — same ledger table, same new pure-weight field; no reshaping needed.
- **Zakat / historical reports** — possible because every transaction and every rate is stored with its own timestamp and never overwritten.

5. Open question resolved — 22K purity factor

The brief lists purity factors as an open question ("the shop's trade practice may not use the textbook ratio"). Asked directly rather than assumed:

- **Value chosen:** 22K gold = 91.66% fine (10g worn → 9.166g pure).
- **Reason:** confirmed by the owner. 22K is 22/24 karat, a repeating decimal (91.666…%) — the owner's figure rounds it down to 91.66%, not up to 91.67%.